PAPER_696: Antennae Galaxies (NGC 4038/4039): Active Merger and Shock-Triggered Star Formation

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Abstract

The Antennae Galaxies (NGC 4038/4039) are the closest major galaxy merger, exhibiting spectacular tidal tails, active starburst regions, and X-ray bright shock ionization zones. UQFF modifies the merger gravity and shock-enhanced star formation.

1. Parameters

Parameter	Value
M_{4038}	$10^{11} M_{\odot}$
M_{4039}	$8 \times 10^{10} M_{\odot}$
Separation	7 kpc
SFR (burst)	20 $M_{\odot}$/yr

2. UQFF Merger Gravity

$$g_{merge} = \frac{G(M_1 + M_2)}{r^2}(1 + f_{TRZ}) + \rho_{shock} V g \frac{\rho_{SCm}}{\rho_{UA}}$$

3. Dynamical Friction (Chandrasekhar)

$$t_{df} = \frac{1.17\, v^3}{G^2 M_{tot}\, \rho_{avg}\, \ln \Lambda}$$

4. UQFF Star Formation Rate

$$\mathrm{SFR}_{UQFF} = \mathrm{SFR}_{burst} \frac{\rho_{shock}}{\rho_{UA}} \left(1 + \frac{\rho_{SCm}}{\rho_{UA}}\right)$$

5. Binding Energy

$$E_{bind} = -\frac{G M_1 M_2}{d_{sep}}$$

References

- Hibbard et al. (2001), AJ, 122, 2969
- UQFF Framework, Star-Magic Session 174

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